OPERATIONAL AMPLIFIER DESIGN

Karapatsias Nikolaos 10661

Requirements to meet:

Parameter	Requirement
A(0)	>=30 dB
BW	>= 500 Khz
PM	PM >=45 deg
Isup	<= 50 uA
Voffset	< 20 mV
SL	>= 1 V/ms

Theoretical design:

Step 1(Choosing L)

Firstly we are gonna choose the length of the transistors. Intuitively we assume that 45nm is too low, so as a starting point we set an arbitrary length of **L=180nm** (4 times bigger than the technology).

Step 2(Choosing miller capacity)

To ensure a phase margin greater than 45 we need to ensure the miller capacitor is meeting the Cc>0.22*CL hence Cc>0.88 pF. We are arbitrarily going to choose a fairly larger value of **Cc = 1 pF**. Keep in mind that we may need to increase this value later on if needed to meet the phase margin and slew rate constraints.

Step 3(Setting current I5)

Given the slew rate and the miller capacitance we can easily calculate the current of I5 using the formula $I5 = \frac{Slew\ rate}{Cc} = 1nA$. Although this value is huge for our design (we need max 50uA for the whole system). So we will try to restrict the current our differential pair M1-M2 consumes arbitrarily at 1uA each, which yields I5 = I_{d1}+I_{d2} => **I5 = 2uA.** We might need to change this value later on too.

Step 4($W_{1,2,3,4}$)

The transconductance of the transistors on differential pair they can be calculated by the formula $Gm = GB \cdot Cc$. Our requirement is bandwidth being greater than 500 Khz so we arbitrarily pick a x6 bigger value to add a safety margin and we may need to change that later on if needed. So the transconductance of M1 and M2 is $Gm_{1,2}$ = 500*2π*6*1 = 18,8 ≈ 19 uS.

Now we can calculate the area of our differential pair M1 and M2 by using the formula:

S1 = S2 = $\frac{Gm1,2^2}{K2' \cdot I1,2} = \frac{19^2}{2 \cdot 154 \cdot 1} = 1.3$ So we approximately we chose a ratio of 2 for S1 and S2.Therefore, since L = 180 then **W_1 = W_2 = 2*180 = 360 nm.**

For simplicity reasons as a starting point we are going to choose the same area for M3 and M4 transistors hence **W_3 = W_4 = 2*180 = 360 nm.**

Step 5(W_5)

To calculate the area of M5 we first need to calculate the V_{ds5sat} by the formula:

$Vds5sat = Vin(min) - Vss - (\sqrt{\frac{I5}{b1}}) - Vth1 = Vin(min) - Vss - (\sqrt{\frac{I5}{\mu nCox\frac{w1}{L}}}) - Vth1 =$

$0.8 - 0 - (\sqrt{\frac{2}{154.5 \cdot 2}}) - 0.585 = 0.8 - 0.09 - 0.585 = 0.125mV$

And therefore $S5 = \frac{2 \cdot I5}{K' \cdot Vds5sat^2} = \frac{2 \cdot 2}{154.5 \cdot 0.125^2} = 4.1 \approx 4$ hence W5 = 4*180 =>

W5 = 720nm

Step 6(W_6)

Since gm2 = 19uS , we can calculate the transconductance of M6 as from the formula: Gm6 = $2.2 \cdot gm2\frac{Cl}{Cc} = 2.2 \cdot 19\frac{4}{1}$ = 167.2 uS. In order to calculate the S6 we need the transconductance of M4 and since M4 and M2 have the same current and W/L we can use the below formula: $\frac{Gm4}{Gm2} = \frac{\sqrt{2 \cdot \mu Coxpmos \cdot \frac{w}{L} Id}}{\sqrt{2 \cdot \mu Coxnmos \cdot \frac{w}{L} Id}} = 0.82$ and therefore Gm4 = Gm2*0.82 = 15.7uS. So now we are ready to calculate the area of M6: S6 = $S4\frac{Gm6}{Gm4} = 2 \cdot \frac{167.2}{15.7} = 21.2 \approx 20$. Hence **W6 = 180*20.2 = 3.6um**.

Note that the current I6 is: I6$\frac{Gm6^2}{2 \cdot K6 \cdot S6} = \frac{167.2^2}{2 \cdot 106.1 \cdot 20} = 6.5uA$

Step 7(W_7)

Now we can calculate the area of M7 by the formula: S7 = $\frac{I6}{I5}S5 = \frac{6.5}{2}4$ = 13 and we are going to round it up to 10, to form a 1 by 2 relation between S6 and S7 hence **W7 = L*10 = 180*10 = 1.8um**

Step 8(W_8)

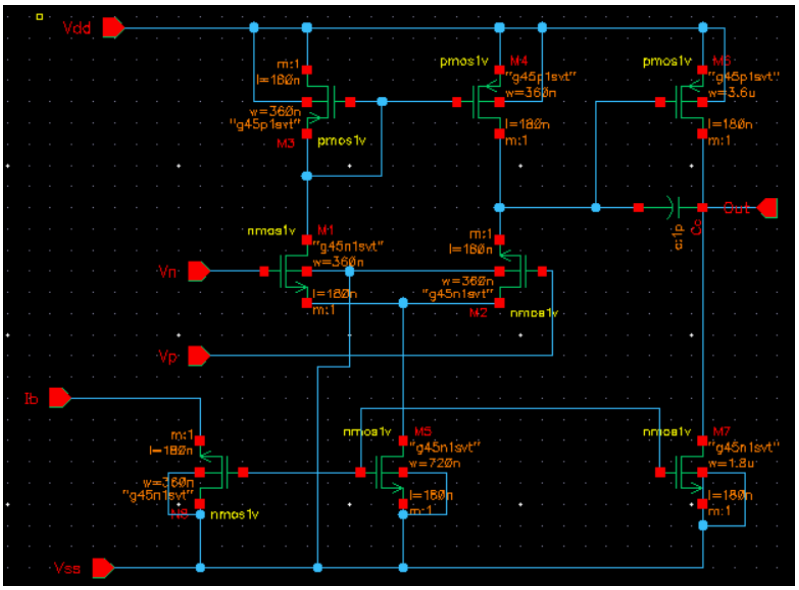
We arbitrarily set W8 of M8 matching the W of our differential input at **W8 = 360nm**.

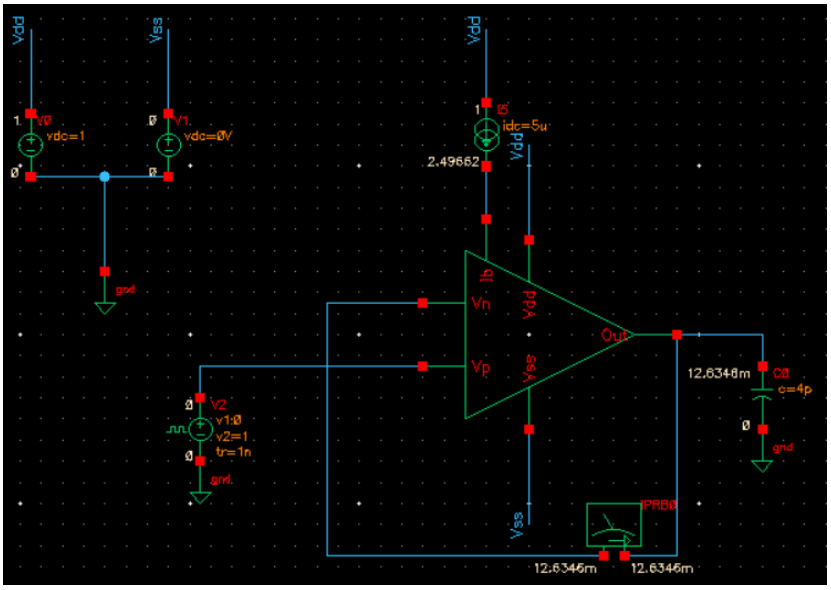
Verdict:

So the final parameters as well as the circuit and the testbench are shown below:

	M1	M2	M3	M4	M5	M6	M7	M8
W (nm)	360	360	360	360	720	3600	1800	360
L (nm)	180	180	180	180	180	180	180	180

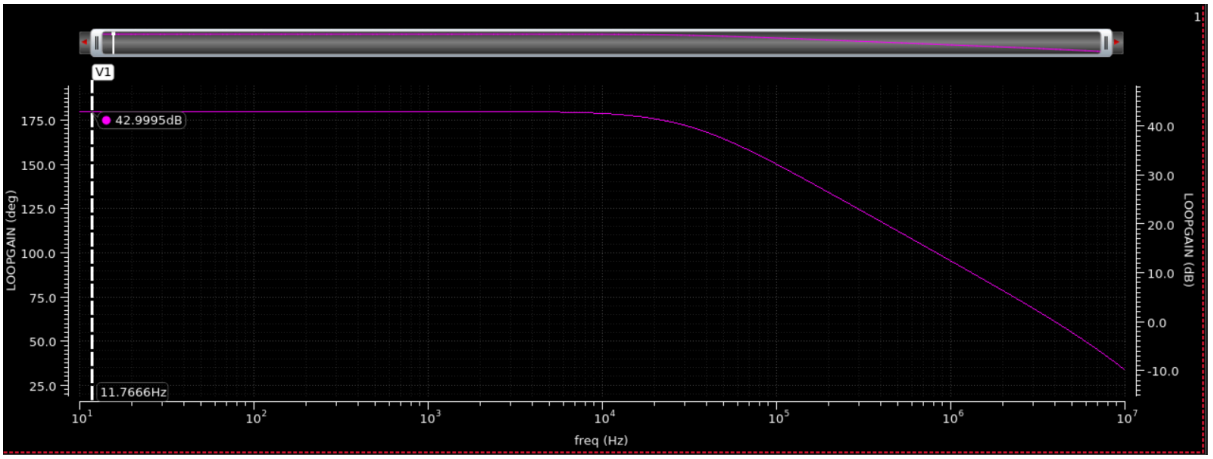
Also:
~ Cc = 2 pF
~ Ib = 5 uA
~ Vdd = 1 V
~ Vdiff = 0.8 V



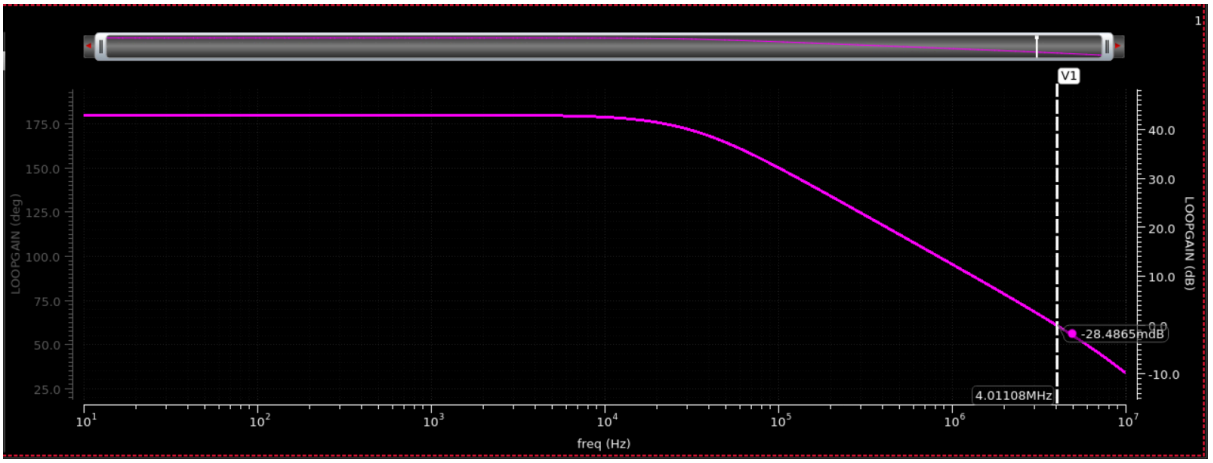


Tests:

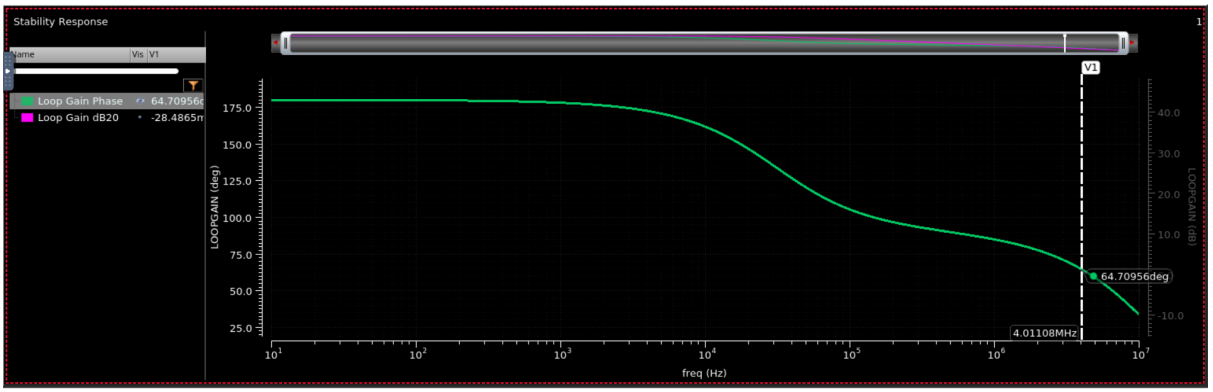
~ Gain A = 43 dB



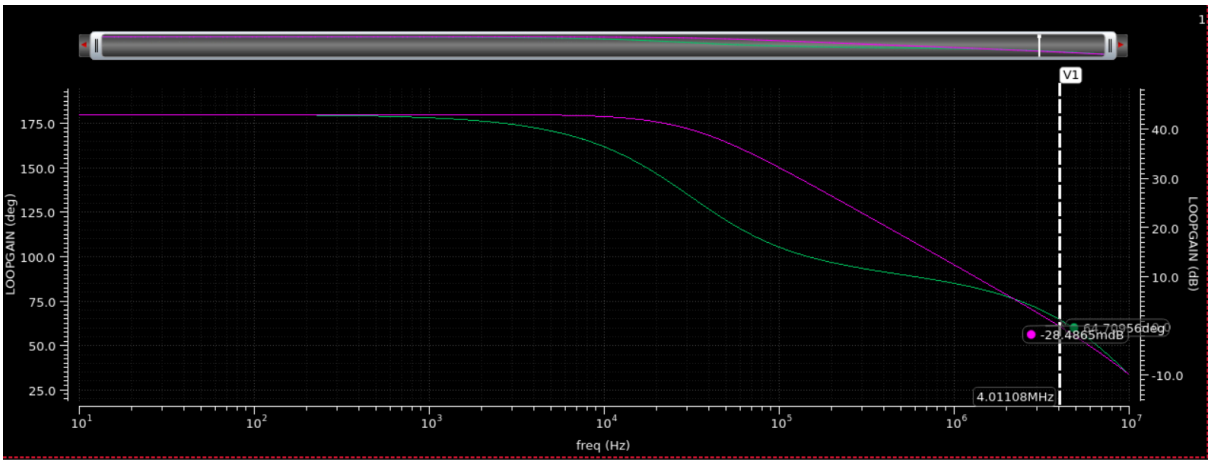
~ **Bandwidth** is around **4 Mhz**:



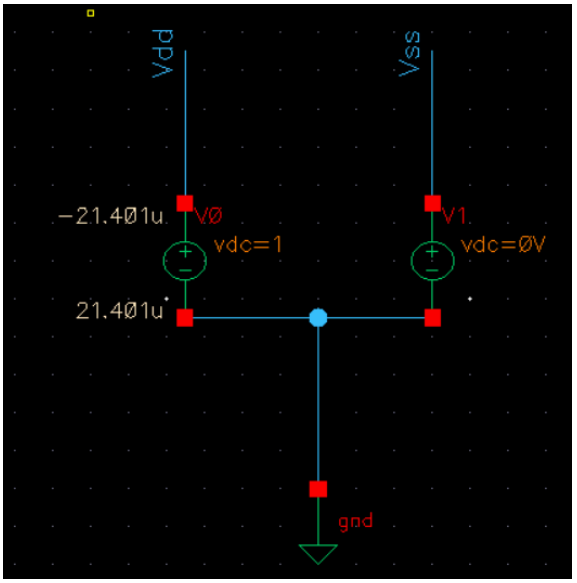
~ **PM = 64.7 deg**:



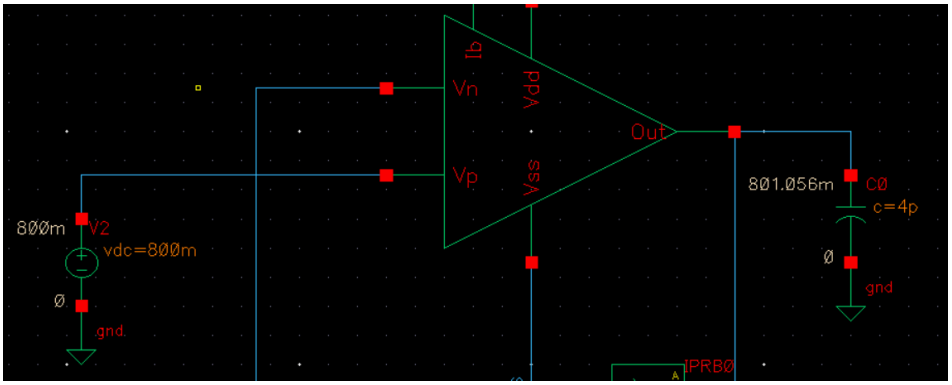
And an **overall view**:



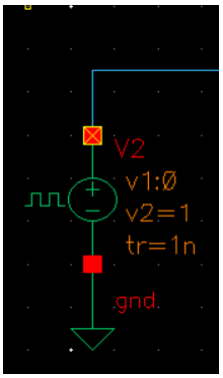
~ Isupp = 21.4 uA



~ Offset = |800-801.05| => **Offset = 1.05mV**:



~ We replace V2 with a pulse generator to calculate the **slew rate = 4.5 V/us**:

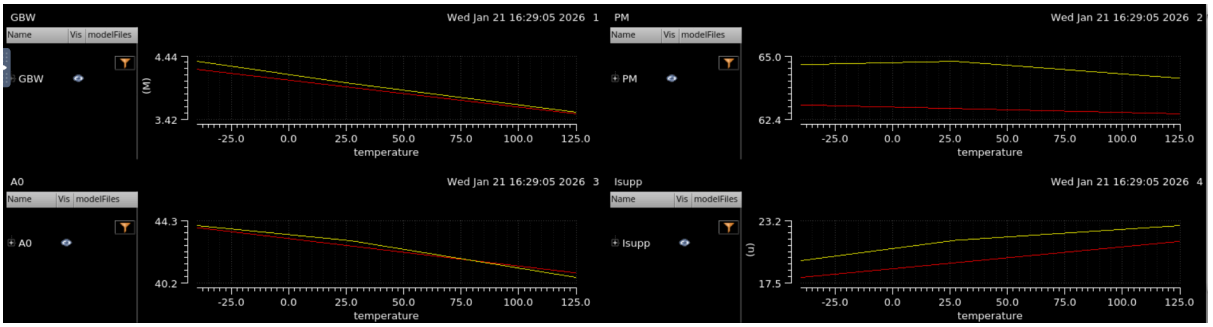


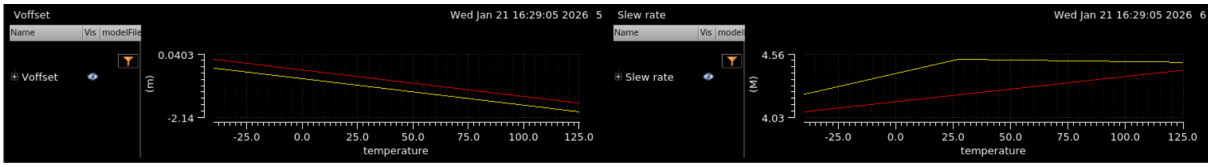
~More accurately we can see the results provided by virtuoso:

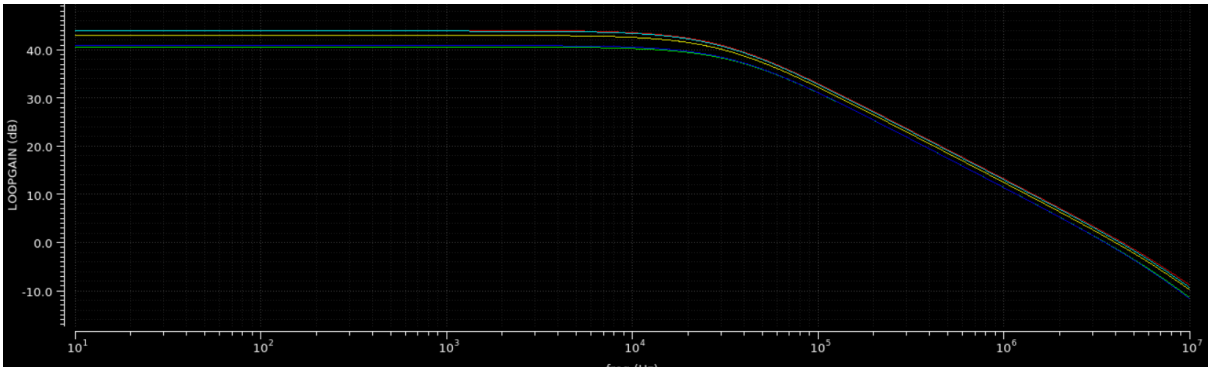
GBW	expr	getData("phaseMarginFreq" ?result "stb_margin")	3.999M	✓		
PM	expr	getData("phaseMargin" ?result "stb_margin")	64.78	✓		
A0	expr	value(db(mag(getData("loopGain" ?result "stb"))) 1)	43	✓		
Isupp	expr	IDC("/V0/MINUS")	21.4u	✓		
Voffset	expr	(VDC("/V4/PLUS") - VDC("/I0/Out"))	-1.056m	✓		
Slew rate	expr	ymax(deriv(vtime('tran "/net4")))	4.517M	✓		

~Last but not least we are running a **corner benchmark** for ff and ss for temperatures from -40 up to 145. The results are shown below:

Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	FF_0	FF_1	SS_0	SS_1
Op_amp_Op_a...	GBW	3.999M				3.504M	4.359M	4.227M	3.504M	4.359M	3.531M
Op_amp_Op_a...	PM	64.78				62.58	64.78	62.95	62.58	64.64	64.07
Op_amp_Op_a...	A0	43				40.58	44	43.87	40.87	44	40.58
Op_amp_Op_a...	Isupp	21.4u				17.98u	22.77u	17.98u	21.31u	19.53u	22.77u
Op_amp_Op_a...	Voffset	-1.056m				-1.956m	-141.1u	-141.1u	-1.655m	-445.7u	-1.956m
Op_amp_Op_a...	Slew rate	4.517M				4.071M	4.517M	4.071M	4.425M	4.22M	4.491M





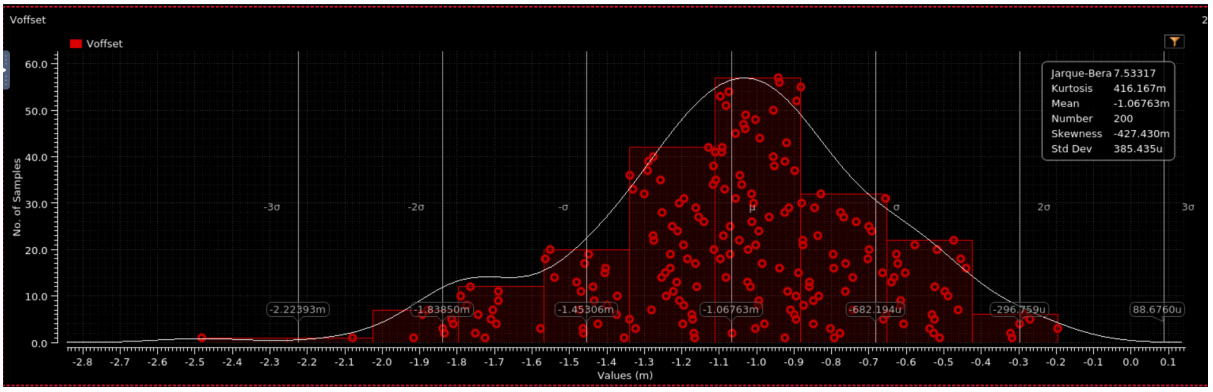


PVT Results

A/A	Minimum	Nominal	Maximum	Specifications
A(0)	40.5 dB	43 dB	44 dB	>= 30dB
BW	3.5 MHz	4 MHz	4.36 Mhz	>= 500Khz
PM	62.58 deg	64.7 deg	64.78 deg	>= 45 deg
Isup	17.98 uA	21.4 uA	22.77 uA	<= 50 uA
Voffset	1.95 mV	1.05 mV	0.141 mV	< 20 mV
SL	4.1 V/us	1.8 V/us	4.5 V/ms	>= 1V/ms

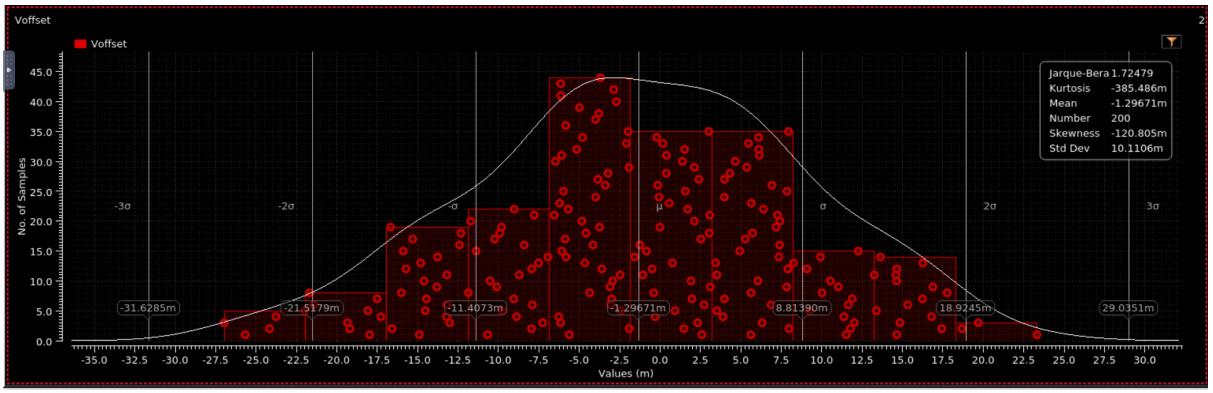
<u>Monte carlo test</u>:

~ Process distribution



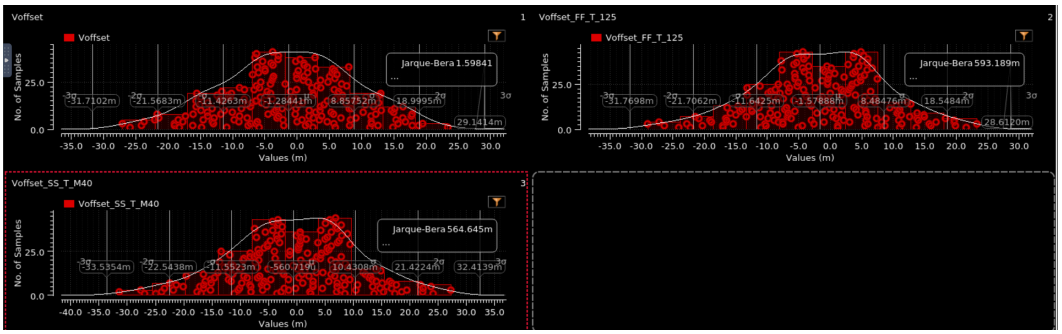
Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 100 %(200 passed/200 pts)	Confidence Level: <not set>	Filter: <not set>							
	Op_amp_Op_amp_tb_1								
	GBW(summary)	0% (0/200)	0		0	0	0		200
	GBW	0% (0/200)	0	info	0	0	0		200
	PM(summary)	0% (0/200)	0		0	0	0		200
	PM	0% (0/200)	0	info	0	0	0		200
	A0(summary)	0% (0/200)	0		0	0	0		200
	A0	0% (0/200)	0	info	0	0	0		200
	Isupp(summary)	100% (200/200)	20.3u		22.54u	21.41u	453.3n		0
	Isupp	100% (200/200)	20.3u	info	22.54u	21.41u	453.3n		0
	Voffset(summary)	100% (200/200)	-2.482m		-195.8u	-1.068m	385.4u		0
	Voffset	100% (200/200)	-2.482m	info	-195.8u	-1.068m	385.4u		0
	Slew rate(summary)	0% (0/200)	0		0	0	0		200
	Slew rate	0% (0/200)	0	info	0	0	0		200

~ Process and mismatch distribution



Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 100 %(200 passed/200 pts)	Confidence Level: <not set>	Filter: <not set>							
	Op_amp_Op_amp_tb_1								
	GBW(summary)	0% (0/200)	0		0	0	0		200
	GBW	0% (0/200)	0	info	0	0	0		200
	PM(summary)	0% (0/200)	0		0	0	0		200
	PM	0% (0/200)	0	info	0	0	0		200
	A0(summary)	0% (0/200)	0		0	0	0		200
	A0	0% (0/200)	0	info	0	0	0		200
	Isupp(summary)	100% (200/200)	18.54u		24.55u	21.44u	1.035u		0
	Isupp	100% (200/200)	18.54u	info	24.55u	21.44u	1.035u		0
	Voffset(summary)	100% (200/200)	-26.98m		23.32m	-1.297m	10.11m		0
	Voffset	100% (200/200)	-26.98m	info	23.32m	-1.297m	10.11m		0
	Slew rate(summary)	0% (0/200)	0		0	0	0		200
	Slew rate	0% (0/200)	0	info	0	0	0		200

~ Mismatch distribution



Test	Name	Yield	Min	Target	Max	Mean	Std Dev	Cpk	Errors
Yield Estimate: 100 %(200 passed/200 pts)	Confidence Level: <not set>	Filter: <not set>							
	Op_amp_Op_amp_tb_1								
	Voffset(summary)	100% (200/200)	-31.42m		27.26m	-1.141m	10.99m		0
	Voffset	100% (200/200)	-27.1m	info	23.35m	-1.284m	10.14m		0
	Voffset_FF_T_125	100% (200/200)	-29.03m	info	23.3m	-1.579m	10.06m		0
	Voffset_SS_T_M40	100% (200/200)	-31.42m	info	27.26m	-560.7u	10.99m		0

Monte Carlo results

A/A	Mean (mV)	Standard deviation (mV)
Process	1.068	385.4
Process+Missmatch(2)	1.297	10.1
Mismatch (tt 25)	1.284	10.14
Mismatch (ff 125)	1.579	10.06
Mismatch (ss -40)	0.56	10.99